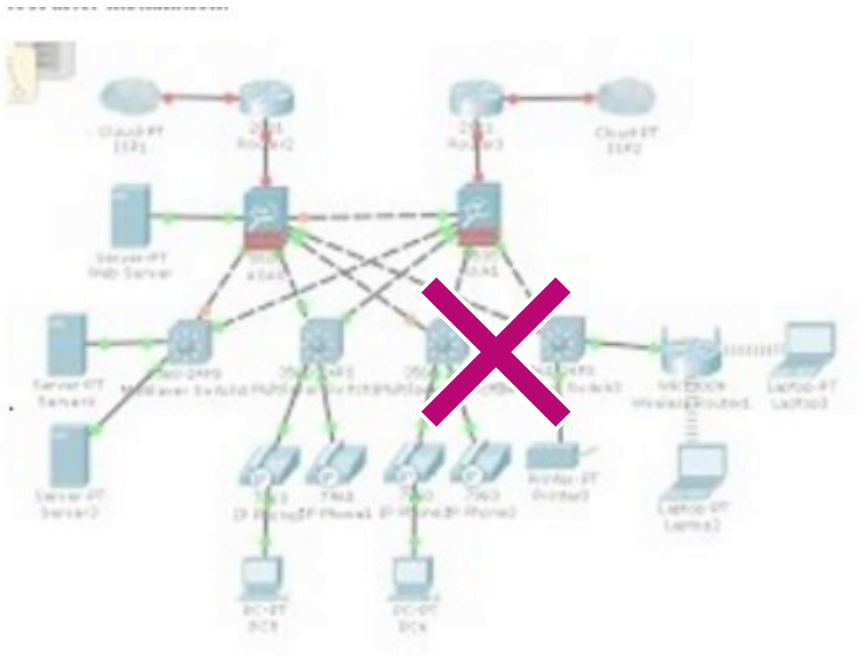
Mini-Project: Setting Up a Networking Simulation Tool

1. **Introduction**
Networking simulation tools enable users to design, configure, and test network environments without physical hardware. This project utilizes Cisco Packet Tracer to create a simple network topology, configure devices, and test connectivity.

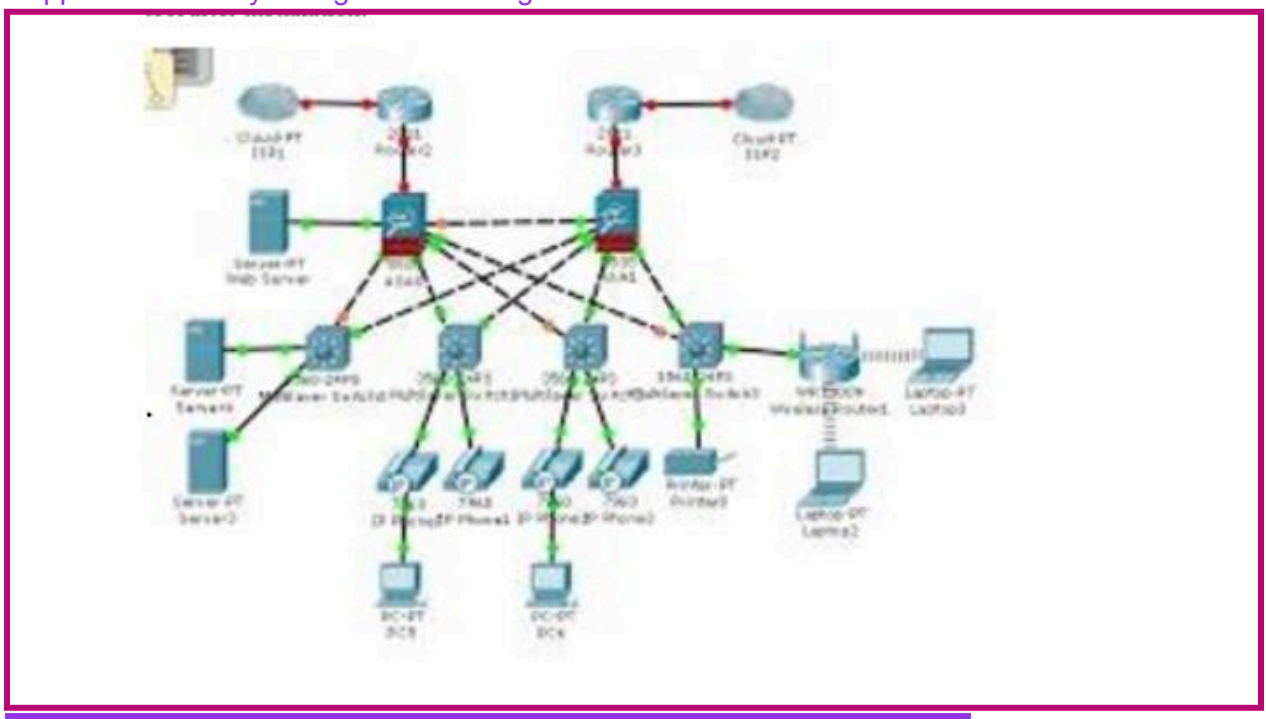
2. **Simulation Tool**
- Cisco Packet Tracer: A powerful network simulation tool used for configuring and testing network setups.

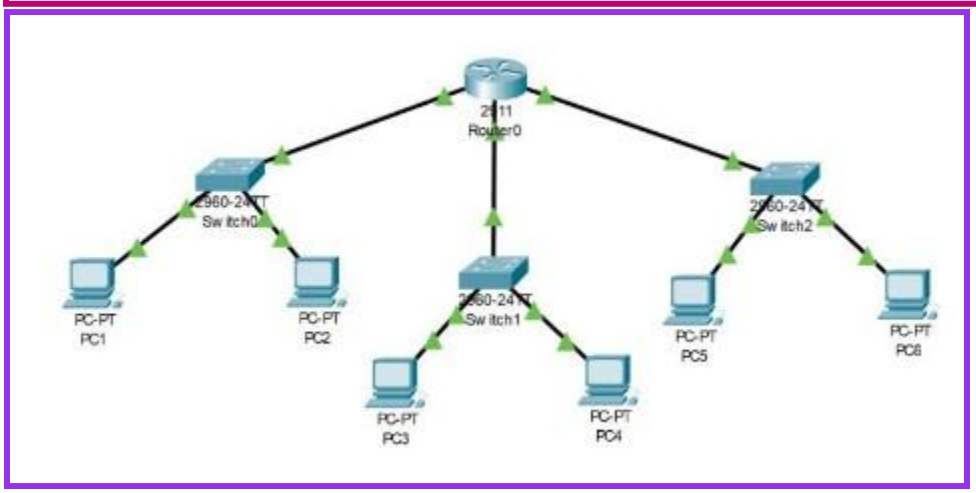
3. **Tool Installation**
Cisco Packet Tracer was successfully installed and launched. Below is a screenshot showing the tool in operation.



4. **Network Design and Configuration**
The network topology was created using the following steps:
- Added two PCs and a switch to the workspace.
- Connected devices using appropriate cables.
- Assigned IP addresses.

- Applied necessary configuration settings.





5. **Connectivity Test**
The 'ping' command was used to verify connectivity. A successful test result confirmed proper communication between devices.

6. **Project Documentation**
The project was saved for future reference.

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Date, 2I150/0063/2025

1. **Objective:** Configure IP addresses and VLANs in a virtual network to enhance segmentation, security, and performance.

2. **Lab Setup**

Tools: Cisco Packet Tracer

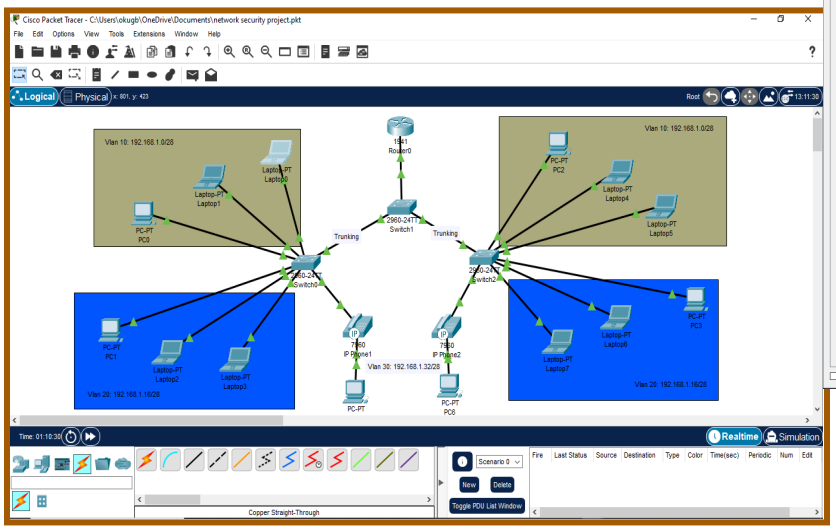
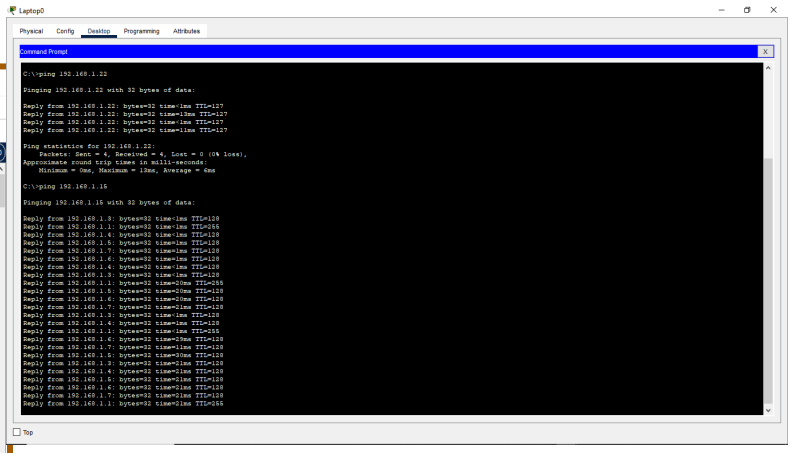
3. **Devices:** 1 Router (Inter-VLAN routing), 1 Switch (VLAN assignment)
   i. **VLAN 10 (Admin_Traffic) Devices:** PC1: 192.168.1.2/28, PC2: 192.168.1.3/28, PC3: 192.168.1.4/28
   ii. **VLAN 20 (Guest_Traffic) Devices:** PC4: 192.168.1.18/28, PC5: 192.168.1.19/28, PC6: 192.168.1.20/28

4. **Device IP Configuration**

- **VLAN 10 PCs:** Default Gateway: 192.168.1.1
- **VLAN 20 PCs:** Default Gateway: 192.168.1.17

5. **Verification**

v **Ping Tests:**

- **PC1 (192.168.1.2) → PC2 (192.168.1.3):** ✅ Success
- **PC4 (192.168.1.18) → PC5 (192.168.1.19):** ✅ Success
- **PC1 (192.168.1.2) → PC4 (192.168.1.18):** ✅ Success

6. **Conclusion:** Successfully configured VLAN segmentation with proper IP addressing and inter-VLAN routing. All devices can communicate within and across VLANs as intended.